ECx00x&EG800K&EG810M&EG91xN&EG915K&EG950A Series

Low Power Mode Application Note

LTE Standard Module Series

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Quectel Wireless Solutions Co., Ltd.
Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China
Tel: +86 21 5108 6236
Email: info@quectel.com

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About the Document

Revision History

Version	Date	Author	Description
-	2022-03-14	Evin ZHU/ Chaim QIAN	Creation of the document
1.0	2023-10-18	Evin ZHU/ Chaim QIAN	First official release
1.1	2024-12-12	Marvin NING	Updated the applicable modules: ● Added EG915K-EU, EC800M-CN, EC800K-CN and EC600M series. ● Deleted EOL products EC200S series and EG912Y-EU. ● Updated EG810M-EU to EG810M series.

Contents

About the Document **3**
Contents **4**
Table Index **5**
Figure Index **6**

1 Introduction **7**
1.1. Applicable Modules 7

2 Related Hardware Interfaces **8**
2.1. Communication Interfaces 8
2.2. Power Management Pins 8

3 URC Introduction **9**
3.1. Set URC Port 9
3.2. Manage URC Event Report 10

4 Implementation of Low-Power Solutions **11**
4.1. Low-Power Solution Based on the UART Interface 11
4.1.1. Hardware Connection 11
4.1.2. Software Initialization 12
4.1.3. Implementation Method 12
4.2. Low-Power Solution Based on USB Suspend/Resume 13
4.2.1. Hardware Connection 13
4.2.2. Software Initialization 13
4.2.3. Implementation Method 13

5 Software Mechanisms **14**
5.1. Module Sleep Mechanism 14
5.2. Module Sleep Process 14
5.3. USB Sleep Mechanism 15
5.4. Host Wake-Up 15
5.4.1. Host Wake-Up Event 15
5.4.2. Host Wake-Up Event Setting 15
5.5. Module Wake-Up 15

6 Appendix References **16**

Table Index

Table 1: Applicable Modules 7
Table 2: Power Management Pins 8
Table 3: Related Documents 16
Table 4: Terms and Abbreviations 16

Figure Index

Figure 1: Connection Diagram Based on UART 11
Figure 2: Connection Diagram Based on USB Suspend/Resume 13
Figure 3: Module Sleep Process 14

1 Introduction

In embedded applications, the power consumption of products is increasingly valued by users, which is especially important for battery-powered devices. Quectel provides power management solutions on reducing product power consumption and extending battery life to reduce product operating costs and improve product competitiveness. The power consumption can be managed by software or hardware design. This document focuses on how to minimize power consumption through software design.

This document describes peripheral hardware interfaces, URC, low-power implement solutions, and the sleep and wake-up software mechanism of the Quectel ECx00x family, EG800K series, EG810M series, EG91xN family, EG915K-EU and EG950A series modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00x	EC200A Series
	EC200N-LA
	EC600M Series
	EC800M-CN
	EC800K-CN
-	EG800K Series
-	EG810M Series
EG91xN	EG912N-EN
	EG915N Series
-	EG915K-EU
-	EG950A Series

2 Related Hardware Interfaces

2.1. Communication Interfaces

The communication between the module and the host system is realized through USB and UART interfaces.

2.2. Power Management Pins

There are some other functional pins provided for the host system to achieve the module's power management.

Table 2: Power Management Pins

Pin Name	I/O	Description
MAIN_DTR	IN	This pin is used for the host to wake up the module from sleep mode or allow the module to enter the sleep mode. When MAIN_DTR is at a high level, the host allows the module to enter the sleep mode. When the module is in the sleep mode, driving MAIN_DTR low will wake up the module.
MAIN_RI	OUT	This pin is used by the module to notify the host that there is a URC event that needs to be reported. When there is no URC event, MAIN_RI keeps at a high level. When there is a URC event to report, MAIN_RI outputs a low pulse. At this time, if the host is in sleep mode, it will be awakened to process the reported URC.

NOTE

1. When MAIN_DTR is at a high level, whether the module enters the sleep mode also depends on other conditions. If MAIN_DTR is at a low level, the module cannot enter the sleep mode.
2. URC can be reported through the UART interface. When there is a URC to report, MAIN_RI generates a low pulse to notify the host, and then the URC content will be output.
3. For detailed interface and pin information of a particular module, refer to the corresponding Hardware Design.

3 URC Introduction

In the interaction between a module and a host, the module not only responds to requests from the host but also actively informs the host of external events, such as incoming calls and SMS. In general, the information actively reported by the module is called URC (Unsolicited Result Code).

The module's processing of URCs includes two aspects: reporting event information to the host actively and generating a low pulse on the MAIN_RI pin. The MAIN_RI pin informs a sleeping host of an incoming URC in the form of a pulse. Because the module supports several types of URCs, it may wake up the host too frequently and reduce the power management efficiency of the whole system. To avoid frequent reporting of URCs, the host should select needed URCs based on its own needs.

3.1. Set URC Port

Generally, the module communicates with the host by either the UART or the USB interface, and so does the reporting of URCs.

The URC-reporting port can be configured by **AT+QURCCFG**. For more details about the command, refer to ***document [1]***.

3.2. Manage URC Event Report

The module supports various types of URCs, such as the network status URC used to report the module's current network status, the SMS URC used to report the new incoming short messages, and the ring URC used to report an incoming call.

You can enable or disable the reporting of most URCs by **AT+QINDCFG**. Thus, the host can enable or disable the reporting of URCs according to actual needs.

Two examples are provided below.

- When the host system starts for the first time, enable the reporting of URCs related to network, SMS, and phone calls.
- Before the host enters the sleep mode, disable the reporting of part of network-related URCs and enable the reporting of URCs related to SMS and phone calls only.

In this way, the power management efficiency of the whole system is improved and the basic requirements of the host are met.

NOTE

About how to set the URC report by **AT+QINDCFG**, refer to ***document [1]***.

4 Implementation of Low-Power Solutions

4.1. Low-Power Solution Based on the UART Interface

The host communicates with the module through the UART interface. This chapter describes the specific implementation methods of the low-power solution based on the UART interface.

4.1.1. Hardware Connection

- The host is connected to the module through the UART interface.
- The host I/O pin is connected to the MAIN_RI pin of the module, and the used host I/O pin needs to support the interrupt wake-up function.
- The host I/O pin is connected to the MAIN_DTR pin of the module.

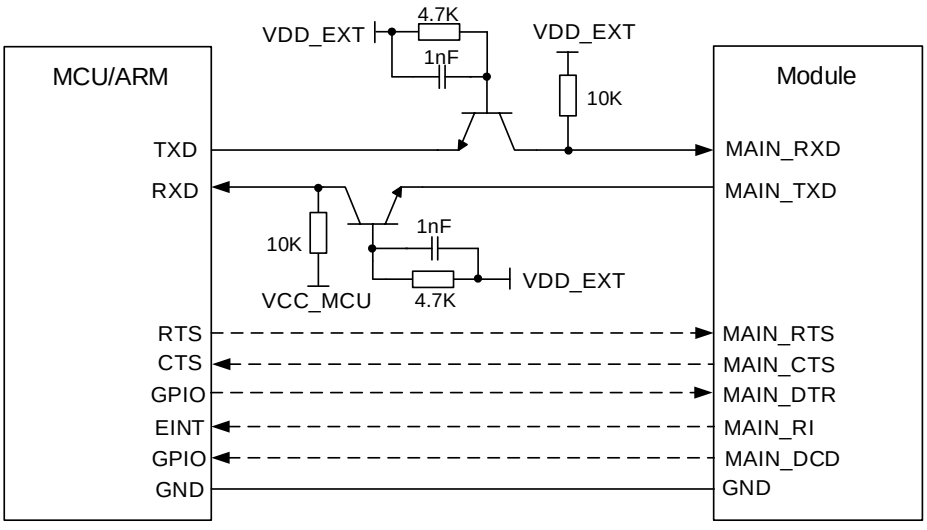


Figure 1: Connection Diagram Based on UART

The UART interface is an important channel for the communication between the module and the host. By multiplexing the DTR as a common GPIO, the host can control the sleep mode of the module to manage the power consumption.

When the host's DTR outputs a high level, the module is allowed to enter sleep mode. When the host's DTR outputs a low level, the module is forbidden to enter sleep mode if it is in a wake-up state or will be awakened if it is in sleep mode.

NOTE

For the design of circuits shown in dotted lines in the above figure, please refer to that shown in solid lines, and pay attention to the direction of connection.

4.1.2. Software Initialization

- **AT+QSCLK=1**

  Allow the module to enter the sleep mode (whether the module enters the sleep mode also depends on other conditions). For more details about the command, refer to ***document [1]***.

- **AT+QURCCFG="urcport","uart1"**

  Specify the UART interface as the URC-reporting port. For more details about the command, refer to ***document [1]***.

4.1.3. Implementation Method

- The module enters the sleep mode:

  The host pulls up the MAIN_DTR pin of the module to enable the module to enter sleep mode.

- The module wakes up from the sleep mode:

  The host pulls down the MAIN_DTR pin of the module to control the module's exit from sleep mode.

- The module wakes up the host:

  When the module has an active event URC to report, the MAIN_RI pin of the module generates a low pulse (120 ms duration by default) to inform the host of the incoming URC. You need to ensure that the host can be woken up by the level change of MAIN_RI when it is in a sleep state.

4.2. Low-Power Solution Based on USB Suspend/Resume

For this low-power solution, the host is required to support USB suspend/resume. This chapter describes the specific implementation methods of the low-power solution based on USB suspend/resume.

4.2.1. Hardware Connection

The host communicates with the module through the USB interface, as shown by the figure below.

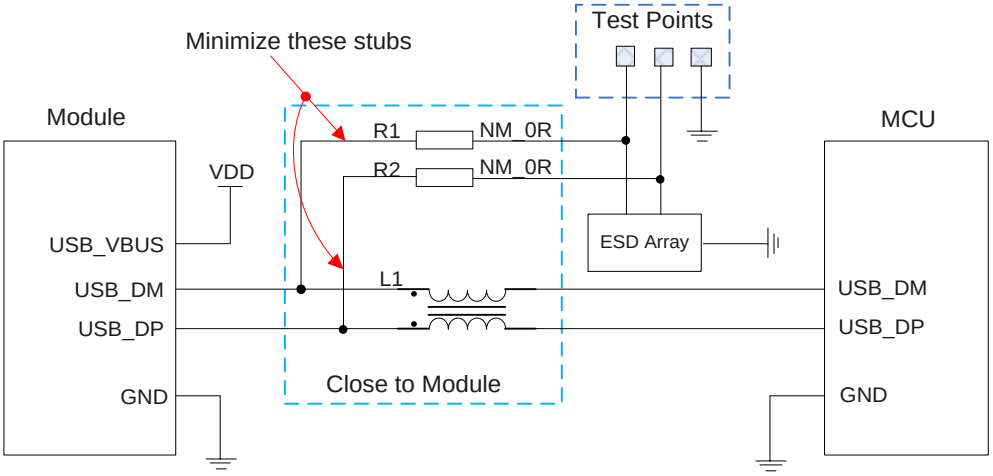


Figure 2: Connection Diagram Based on USB Suspend/Resume

4.2.2. Software Initialization

- **AT+QSCLK=1**

  Allow the module to enter the sleep mode (whether the module enters the sleep mode also depends on other conditions). For more details about the command, refer to ***document [1]***.

4.2.3. Implementation Method

- The module enters the sleep mode:

  The host system enters the sleep state and USB is suspended.

- The module wakes up from the sleep mode:

  The host uses the USB resume functionality to wake up the module.

5 Software Mechanisms

5.1. Module Sleep Mechanism

There is a lower-priority sleep task inside the module, which is used to detect whether the module can enter the sleep mode or not. The sleep control variables (**AT+QSCLK**) and other services (such as RF, USB, and UART) have rights to vote on the sleep task to decide whether or not the module can enter the sleep mode. Only when both the sleep control variables and other service tasks agree the module to enter the sleep mode, the sleep task is executed and the module enters sleep mode.

NOTE

After the module enters sleep mode, the RF task is not stopped and the module works in DRX mode. DRX is a method that saves the module's power consumption and its basic principle is to make the module enter the sleep state periodically.

5.2. Module Sleep Process

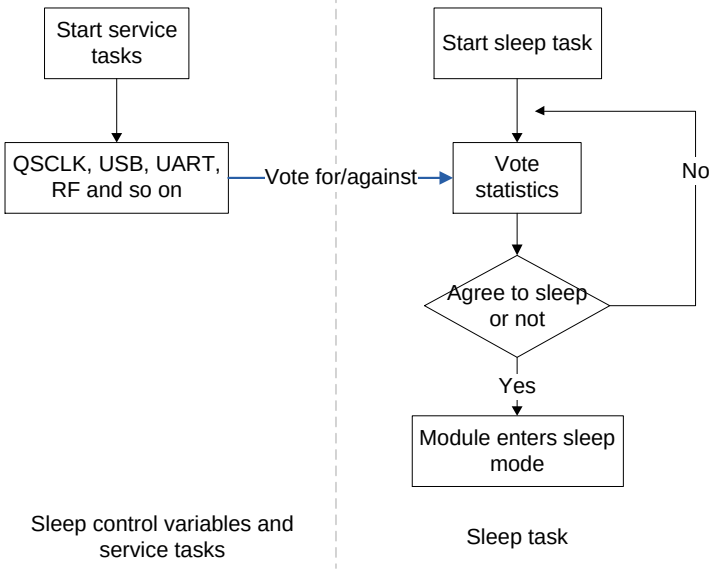


Figure 3: Module Sleep Process

5.3. USB Sleep Mechanism

When USB is working normally, root hub in the hub or HCD sends SOF package periodically (full-speed USB sends one package at an interval of 1 ms, high-speed USB sends one package at an interval of 125 μs). At this time, USB on the module votes against the module to enter sleep mode.

According to the description of suspend in the USB specification, when the host system sets the port that is connected to the USB device in hub or root hub as suspend, hub or root hub stops sending SOF package and the USB enters suspend mode. Then the USB interface on the module enters suspend mode and votes for the module to enter sleep mode.

NOTE

1. The module's USB suspend functionality provided by Quectel supports different host systems. Users can control the module's USB to enter suspend mode on the host system.
2. If you are using a USB driver developed on your own, please make sure that the driver supports both global suspend and selective suspend functionalities.

5.4. Host Wake-Up

5.4.1. Host Wake-Up Event

Host wake-up event is the event that the module takes the initiative to wake up the host by reporting information to the host when the host is in sleep mode. Host wake-up event is also called URC. For more details, please refer to ***Chapter 3***.

5.4.2. Host Wake-Up Event Setting

The host can configure the URC-reporting port by **AT+QURCCFG**, for example, UART or USB AT port. In particular scenarios, the host can specify the URC-reporting port according to the actual hardware connection. The default port is USB AT port.

5.5. Module Wake-Up

The host wakes up the module by pulling down the MAIN_DTR pin or by USB Resume. For more details, refer to ***Chapter 4.1.3*** and ***Chapter 4.2.3***.

6 Appendix References

Table 3: Related Document

Document Name	
[1]	Quectel_ECx00x&EG800K&EG810M&EG91xN&EG915K&EG950A_Series_AT_Commands_Manual

Table 4: Terms and Abbreviations

Abbreviation	Description
DRX	Discontinuous Reception
DTR	Data Terminal Ready
HCD	Host Controller Driver
I/O	Input / Output
PDCCH	Physical Downlink Control Channel
RF	Radio Frequency
RI	Ring Indicator
SOF	Start-of-Frame
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment
URC	Unsolicited Result Codes
USB	Universal Serial Bus